Detecting n-connected maps by their action on truncation and on homotopy groups

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:pik-conn' (source lines 2464-2471, Corollary 8.8.5); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence presenting the corollary as the converse of the corresponding statement about homotopy groups of n-connected maps.

Editorial note. The retained counter values (chapter 8, section 8, theorem offset 4) make the retained corollary print as Corollary 8.8.5, matching the original book numbering; theorem-class environments share one counter through aliascnt.

Editorial note. The three external references are navigation pointers to results of section 8.4 and section 8.8 that the retained proof invokes by label; the counted proof is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: the criterion for n-connected maps by truncation and homotopy groups (Corollary 8.8.5).

Allowed standard prerequisites (editorial).

- n-connected maps. f : A -> B is n-connected if trunc n (fiber_f b) is contractible for every b : B.. The goal after the group-theoretic hypotheses are used is exactly the contractibility of trunc n (fiber_f b) for all b : B.
- The long exact sequence of a fibration. A fiber sequence gives a long exact sequence of homotopy groups, so vanishing and surjectivity statements about pi_k translate into statements about the fiber.. It is applied at source line 2474 to deduce pi_k(fiber) = 0 for k <= n and contractibility of trunc 0 (fiber).
- Truncation does not change low homotopy groups. pi_k(trunc n X) = pi_k(X) for k <= n, and trunc n X is n-connected.. It is the passage at source line 2477 that turns the group information into contractibility via the Whitehead criterion.

Source reference. Original theorem 8.4.8: sources/hott-homotopy/pinned-original-source.tex, source lines 1153–1153. Original result cited by the retained text but

located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 8.4.6: sources/hott-homotopy/pinned-original-source.tex, source lines 1080–1080. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 8.8.4: sources/hott-homotopy/pinned-original-source.tex, source lines 2455–2455. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

As an application, we can deduce the converse of original theorem 8.4.8.

Corollary 8.8.5. *For $n \geq 0$, a map $f : A \to B$ is n-connected if and only if the following all hold:*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ *is an isomorphism.*
(ii) For any $a : A$ and $k \leq n$, the map $\pi_k(f) : \pi_k(A, a) \to \pi_k(B, f(a))$ is an isomorphism.
(iii) For any $a : A$, the map $\pi_{n+1}(f) : \pi_{n+1}(A, a) \to \pi_{n+1}(B, f(a))$ is surjective.

Proof. The "only if" direction is original theorem 8.4.8. Conversely, by the long exact sequence of a fibration (original theorem 8.4.6), the hypotheses imply that $\pi_k(\mathsf{fib}_f(f(a))) = 0$ for all $k \leq n$ and $a : A$, and that $\|\mathsf{fib}_f(f(a))\|_0$ is contractible. Since $\pi_k(\mathsf{fib}_f(f(a))) = \pi_k(\|\mathsf{fib}_f(f(a))\|_n)$ for $k \leq n$, and $\|\mathsf{fib}_f(f(a))\|_n$ is n-connected, by original theorem 8.8.4 it is contractible for any a.

It remains to show that $\|\mathsf{fib}_f(b)\|_n$ is contractible for $b : B$ not necessarily of the form $f(a)$. However, by assumption, there is $x : \|A\|_0$ with $|b|_0 = \|f\|_0(x)$. Since contractibility is a mere proposition, we may assume x is of the form $|a|_0$ for $a : A$, in which case $|b|_0 = \|f\|_0(|a|_0) = |f(a)|_0$, and therefore $\|b = f(a)\|_{-1}$. Again since contractibility is a mere proposition, we may assume $b = f(a)$, and the result follows. □